Latest in Quantum Computing -- diff report

Generated: 2026-08-30T21:32:51.550211 | **Period:** since 2026-08-30 | **Leaves with news:** 9 of 10

Fresh developments: 20 (as_of within 14 days of the window) | **Superseded/conflicting:** 5 | **Background captured:** 33 (older material, first time in the knowledge base)

Reconfirmations in this period (no new information, not shown): 1.

Summary

Leaf	Fresh	Superseded	Conflicting	Background
Academic & theoretical	6	0	0	1
Error correction & logical qubits	3	0	1	8
Industry & funding moves	4	0	0	7
Trapped ion qubits	1	2	0	1
Neutral atom qubits	3	0	0	3
Superconducting qubits	0	2	0	5
Applications & benchmarking claims	2	0	0	2
Spin qubits	1	0	0	3
Photonic quantum computing	0	0	0	3

Industry, academia & claims

Academic & theoretical (`academic-theoretical`)

New developments (6)

Complexity-theoretic and hardness results

- A preprint ("Quantum Speedups Require Structure or Depth," submitted August 19, 2026, not yet peer-reviewed) shows that every t-query, d-round quantum algorithm can be simulated on most inputs with $t^{O(d^2)}$ classical queries — implying that for UNSTRUCTURED problems, a superpolynomial quantum speedup requires super-constant circuit depth, and an exponential speedup requires polynomial depth. (arxiv.org — 2608.19158, as of 2026-08-19, confidence: medium, source: ingest)
- *A new complexity-theoretic result narrowing where quantum speedups can exist, distinct from existing entries in this subtopic.*

New or improved quantum algorithms

- Researchers at the Chinese Academy of Sciences published (Quantum Science and Technology, peer-reviewed, August 28, 2026) a quantum algorithm for topological data analysis on directed graphs, introducing a path-homology approach with a universal digraph-encoding protocol; theoretically guarantees a speedup dependent on data access, with exponential gains possible under certain conditions. (quantumzeitgeist.com — chinese sciences quantum algorithm speeds up, as of 2026-08-28, confidence: medium, source: ingest)

- *A genuinely new peer-reviewed quantum algorithm (path-homology TDA on directed graphs) not previously present in the knowledge base.*
- A peer-reviewed quantum algorithm solves the Sylvester equation A X + X B = C (with block-encoding access to the matrices) by constructing a block-encoding of the solution matrix X, achieving query and gate complexities almost linear in a condition number depending on A and B and logarithmic in dimension and inverse error; it can solve BQP-complete problems efficiently. (Quantum algorithm for linear matrix equations, as of 2026-08-21, confidence: high, source: grok)
- *A new peer-reviewed quantum algorithm for solving Sylvester matrix equations not previously present in the knowledge base.*
- A 2026 preprint develops a quantum algorithm for solving the dynamics of the nonlinear viscous Burgers equation (and extensions to general fluid equations) via Carleman linearization, padding, and Permutation Matrix Representation (PMR) combined with Linear Combination of Hamiltonian Simulations (LCHS), scaling with the off-diagonal norm rather than the matrix norm. (Quantum algorithm for differential equations via permutation matrix representation with application to the Burgers equation, as of 2026-08-19, confidence: medium, source: grok)
- *A new, not-yet-peer-reviewed quantum algorithm for nonlinear differential equations distinct from existing algorithm entries.*

Noise and error-mitigation theory

- Two new preprints, both not yet peer-reviewed: "Practical Error Suppression and Mitigation for Reliable Quantum Computing" (August 20, 2026) frames error suppression, mitigation, and full error correction as complementary layers of one unified strategy for the transitional NISQ-to-early-fault-tolerance regime; and "Quantum Error Mitigation with Diffusion-Like Models" (August 5, 2026) proposes an AI-assisted mitigation framework modeling decoherence as sequences of weak measurements. (arxiv.org — 2608.20453, as of 2026-08-20, confidence: medium, source: ingest)
- *Two specific new preprints not previously present in the error-mitigation theory section, incremental but new additions.*
- A peer-reviewed paper introduces surrogate-enabled zero-noise extrapolation (S-ZNE) that uses classical learning surrogates to perform ZNE with only constant measurement overhead for families of parameterized quantum circuits, achieving accuracy comparable to conventional ZNE in many scenarios and validated on up to 100-qubit tasks. (Sample-efficient quantum error mitigation via classical learning surrogates, as of 2026-08-20, confidence: high, source: grok)
- *A new, distinct overhead-reduction technique for ZNE specifically, complementing but not restating the existing PEC-overhead-reduction and landscape-surrogate results already in the base.*

Newly captured background -- older material, first time in the knowledge base (1)

Complexity-theoretic and hardness results

- OpenAI reported (August 4, 2026) that an internal model ("Astra") solved 10 longstanding open problems spanning mathematics, quantum complexity theory, and theoretical computer science, including results on quantum parallel repetition and stronger hardness bounds for the closest vector problem. This is a vendor claim about unverified results -- no detailed proofs released and no peer-reviewed validation announced -- and must not be reported as settled mathematics. (thequantuminsider.com — openai says next generation model solved 10 major open problems in quantum complexity mathematics, as of 2026-08-04, confidence: low, source: ingest)
- *New vendor announcement not previously recorded; must be flagged as an unverified claim rather than a confirmed result per baseline evidence-status rules.*

Industry & funding moves (`industry-funding-moves`)

New developments (4)

Government programs and public-sector spend

- The U.S. National Science Foundation announced (August 25, 2026) that Princeton University will lead one of eight major federally-funded Quantum Leap Challenge Institutes, targeting hardware fabrication techniques among other goals. (engineering.princeton.edu — princeton lead major federally funded institute unlock quantum computing challenges, as of 2026-08-25, confidence: medium, source: ingest)
- *New federal program announcement distinct from existing NSF NQVL entry.*
- The Canadian government committed CAD $195 million (~USD $140.2M) to Xanadu's "Project OPTIMISM," funding a 158,000-sq-ft manufacturing facility ("Inception") in Toronto, announced August 28, 2026. (globenewswire.com — xanadu secures cad 195 million and establishes landmark advanced photonics facility, as of 2026-08-28, confidence: medium, source: ingest)
- *New government public-sector spend commitment not previously recorded.*
- NSF awarded $290 million total across eight Quantum Leap Challenge Institutes on August 25, 2026, including renewals (e.g. UChicago QuBBE at $37.5M) and new awards focused on sensing, hardware fabrication, and fault tolerance. (NSF Awards $290M Across Eight Quantum Leap Challenge Institutes; Focuses on Sensing, Hardware Fabrication, and Fault Tolerance - Quantum Computing Report, as of 2026-08-25, confidence: high, source: grok)
- *Adds the aggregate $290M funding figure and per-institute detail beyond the previously recorded Princeton-specific institute announcement.*

Vendor roadmaps and committed timelines

- Diraq published a roadmap white paper ("The Case for Silicon," August 27, 2026) targeting over 2 million physical qubits on a single chip by 2031 at under $1/qubit system cost, and opened a Santa Monica technology hub (August 26, 2026, ~20-person team, first product targeted 2029). (thequantuminsider.com — diraq opens us technology hub santa monica, as of 2026-08-27, confidence: medium, source: ingest)
- *New vendor roadmap and facility announcement not previously recorded.*

Newly captured background -- older material, first time in the knowledge base (7)

National-lab procurements and partnerships

- PsiQuantum signed a $125 million agreement with DARPA under the Quantum Benchmarking Initiative on July 22, 2026, its largest U.S. government award to date. (Quantum computing startup PsiQuantum signs $125 million DARPA deal | Reuters, as of 2026-07-22, confidence: high, source: grok)
- *New federal contract/award not previously recorded for PsiQuantum.*
- IBM committed up to $50 million in quantum compute access for the US Genesis Mission over five years, announced July 22, 2026. (IBM commits $50M in quantum access for US Genesis Mission - IBM Research, as of 2026-07-22, confidence: high, source: grok)
- *New public-sector commitment not previously recorded.*
- HPE and Rigetti announced a partnership to deploy a hybrid quantum-classical system (9-qubit Novera QPU) at the Pittsburgh Supercomputing Center TangleLab testbed, funded by a $5 million NSF grant, with operations targeted for 2027. (HPE, Rigetti partner for hybrid quantum-classical computer deployment at Pittsburgh Supercomputing Center testbed - DCD, as of 2026-07-28, confidence: high, source: grok)
- *New national-lab deployment partnership not previously recorded.*

Private funding rounds and investment

- eleQtron raised a €57 million Series A round in 2026, with a stated €54 million order backlog. (thequantuminsider.com — eleqtron secures e57 million in series a funding round, as of 2026-05-05, confidence: low, source: ingest)

- *New private funding round not previously recorded for this company.*
- Photon Queue raised $4 million in a Seed funding round led by Playground Global in July 2026. (Photon Queue raises $4m in Seed round for quantum memory devices - DCD, as of 2026-07-23, confidence: high, source: grok)
- *New private seed funding round not previously recorded for this company.*
- Qolab secured $54.2 million in Series B funding (including convertibles) led by UC Investments in July 2026 to advance superconducting quantum hardware. (Qolab Secures $54.2M Series B Funding for Quantum Hardware Development, as of 2026-07-02, confidence: high, source: grok)
- *New private Series B funding round not previously recorded for this company.*

Vendor roadmaps and committed timelines

- QuEra unveiled plans for a gigaquop-class fault-tolerant system (over 1,000 logical qubits, 10^-9 error rate) targeted for 2028-2029, following its Libra megaquop system for Amazon Braket in 2028. (QuEra Unveils Gigaquop-Class Fault-Tolerant Roadmap and Invites Organizations to Co-Design Quantum Applications, as of 2026-06-25, confidence: high, source: grok)
- *New vendor roadmap and committed timeline not previously recorded for QuEra.*

Applications & benchmarking claims (`applications-benchmarking`)

New developments (2)

Optimization application claims

- Two Fraunhofer IAF publications (August 17, 2026) propose methods for assessing potential quantum advantage under more realistic physical conditions and across increasing problem sizes -- a methodological/benchmarking-rigor contribution rather than a new application claim itself. (thequantuminsider.com — scientists propose more realistic benchmarks for quantum algorithms, as of 2026-08-17, confidence: medium, source: ingest)
- *This is a new methodological benchmarking proposal distinct from any existing recorded benchmark suite or optimization claim.*

Standard benchmark suites and metrics

- An August 2026 arXiv preprint formalizes CLOPS_h as a holistic speed benchmark for layered, hardware-aware circuits on utility-scale processors, binding speed to independently verified layer-fidelity quality envelopes. (CLOPS: Benchmarking System Speed at Utility Scale, as of 2026-08-18, confidence: high, source: grok)
- *This introduces a new variant metric (CLOPS_h) extending the existing CLOPS metric with fidelity-binding, not previously recorded.*

Newly captured background -- older material, first time in the knowledge base (2)

Finance application claims

- ORIENTOM and softwareQ signed a memorandum of understanding (August 2026) to explore quantum and quantum-inspired computing for financial services, banking, and insurance -- named use cases include derivatives pricing, portfolio optimization, risk management, and machine learning. This is an MOU/exploratory-partnership stage claim, not a benchmarked result. (thequantuminsider.com — orientom softwareq quantum computing finance applications, as of 2026-08-12, confidence: medium, source: ingest)
- *This is a new, previously unrecorded vendor partnership announcement in finance, explicitly flagged as exploratory rather than a benchmarked result.*

Standard benchmark suites and metrics

- A June 2026 arXiv preprint introduces Clifford Volume and Free Fermion Volume as scalable volumetric benchmarks that are classically simulable for verification, form universal gate sets, and serve as algorithmic primitives, validated experimentally on Quantinuum H2-1 achieving Clifford Volume of 34. (Clifford Volume and Free Fermion Volume: Complementary Scalable Benchmarks for Quantum Computers, as of 2026-06-22, confidence: medium, source: grok)
- *This is a genuinely new benchmark methodology and metric family not present in the existing knowledge base.*

Error correction & logical qubits

Error correction & logical qubits (`logical-qubits-error-correction`)

New developments (3)

Magic-state distillation and cultivation

- Silicon Quantum Co, with the Centre of Excellence for Quantum Computation, announced a 'Magic Scroll' technique for magic-state cultivation in register-based quantum architectures, achieving |T⟩-state fidelities as low as 10^{-9} using efficient implementations of colour and bilayer codes (6.6.6 and 4.8.8 configurations) on two-qubit registers, leveraging biased noise to mitigate hook errors. (quantumzeitgeist.com — silicon quantum cultivation volumes boost threefold, as of 2026-08-25, confidence: medium, source: ingest)
- *Genuinely new magic-state cultivation result (Silicon Quantum Co, colour/bilayer codes) not present among the existing QuEra/Harvard/MIT and Rosenfeld et al. entries.*

Real-time decoding and decoder hardware

- An FPGA-based neural-network decoder integrated with a superconducting quantum processor demonstrated real-time surface-code error correction at distance 3 with deterministic closed-loop latency of 550 ns (124 ns for decoding), achieving logical performance comparable to offline MWPM decoding. (Real-time Surface-Code Error Correction Using an FPGA-based Neural-Network Decoder, as of 2026-08-24, confidence: high, source: grok)
- *New neural-network FPGA decoder result with distinct latency figures not covered by existing FPGA/Riverlane/RL-control entries.*
- IonQ demonstrated a real-time QEC decoding pipeline on a single Apple M4 Max CPU handling MegaQuOp-scale workloads with up to 408 logical qubits and over 1 million T gates, incurring <0.3% computational stretch at p_CNOT=10^{-4}. (Real-time decoder for a MegaQuOp quantum computer using a single CPU, as of 2026-08-25, confidence: high, source: grok)
- *New single-CPU real-time decoding result at unprecedented logical-qubit and gate-count scale, not present in existing decoder entries.*

Conflicting -- needs review (1)

Bosonic and cat-qubit encodings

- Alice & Bob reported bit-flip lifetimes of 33 to 60+ minutes on its Galvanic Cat architecture (Helium 2 chip) — an overlapping but not clearly superseding range versus the established 'exceeding one hour' figure. Roadmap targets an early fault-tolerant quantum computer (eFTQC) with 100 logical qubits by 2030, noted for context only. (quantumway.org — alice bob how cat qubits are rewriting the rules of fault tolerant quantum computing, as of undated, confidence: low, source: ingest)
- conflicts with: "[d0001:15] Alice & Bob reported bit-flip stability exceeding one hour on its Galvanic Cat architecture"

- *The 33-to-60+ minute range partially undercuts the established 'exceeding one hour' figure without clearly superseding it, so it is flagged for human resolution rather than silently replacing the established figure.*

Newly captured background -- older material, first time in the knowledge base (8)

Below-threshold and break-even demonstrations

- Quantinuum demonstrated computations with up to 94 error-detected logical qubits and 48 error-corrected logical qubits on its 98-physical-qubit Helios processor, at a 2:1 physical-to-logical encoding ratio via code concatenation, with 99.99% state-preparation-and-measurement fidelity. The encoded computations achieved "beyond break-even" performance: logical error rates at least an order of magnitude lower than the corresponding physical-qubit operations. (thequantuminsider.com — quantinuum researchers demonstrates quantum computations with dozens of protected logical qubits, as of 2026-03-10, confidence: medium, source: ingest)
- *First trapped-ion (Quantinuum Helios) below-threshold/break-even logical qubit result in this leaf's knowledge base; no prior entry covers this vendor or ratio.*
- A joint team from the University of Innsbruck, RWTH Aachen, Forschungszentrum Jülich, and AQT demonstrated (published April 2026) a complete fault-tolerant quantum algorithm — Grover's search on three logical qubits, encoded across eight physical qubits — executed without mid-circuit measurements, using feed-forward control on a trapped-ion processor. (uibk.ac.at — quantum computing without interruptions, as of 2026-04-01, confidence: medium, source: ingest)
- *Novel fault-tolerant algorithm demonstration (measurement-free QEC on trapped ions) with no analogous entry in the existing knowledge base.*

Real-time decoding and decoder hardware

- Google demonstrated reinforcement-learning control unifying calibration and computation on the Willow processor, achieving a 3.5-fold logical-stability improvement and a 20% logical-error reduction, published in Nature, July 2026. (quantumcomputingreport.com — google research stabilizes willow quantum processor using continuous reinforcement learning control layers, as of 2026-07-08, confidence: medium, source: ingest)
- *New peer-reviewed logical-error-rate improvement via RL-based control layer, not previously recorded for Willow.*

Surface-code results

- Refinements to the Google Willow below-threshold surface-code result: an author correction to the underlying Nature paper was published in April 2026, and the real-time decoder's average latency is 63 microseconds at distance 5, sustained up to a million cycles. (nature.com — s41586 024 08449 y, as of 2026-04-01, confidence: medium, source: ingest)
- *Adds two previously-unrecorded concrete details (decoder latency figure and paper's author correction) to the established Willow below-threshold entry, beyond mere reconfirmation.*
- Surface-code logical operations including merge/split, CNOT, Hadamard, and phase gates were experimentally realized on distance-3 rotated surface-code patches using a 107-qubit superconducting processor with multi-round syndrome extraction and neural-network decoding. (Surface code logical operations on a superconducting quantum processor, as of 2026-07-01, confidence: high, source: grok)
- *New demonstration of a full set of logical operations on distance-3 patches, not previously recorded among the below-threshold memory results.*

qLDPC and high-rate codes

- IQM demonstrated "directional tile" qLDPC error-correction codes reducing the per-logical, per-round error rate by 1,000x versus surface codes on existing Crystal hardware, June 2026. (interestingengineering.com — iqm approach reduces qubit overhead 1000 times, as of 2026-06-24, confidence: medium, source: ingest)

- *New vendor/code result (IQM directional-tile qLDPC) with no prior entry in the base.*
- A QuEra/Harvard/MIT collaboration posted a preprint (April 2026) demonstrating a qLDPC code encoding 1,156 logical qubits into 2,304 physical qubits (a 2:1 physical-to-logical ratio), achieving a per-logical-qubit, per-round error rate of approximately 1.3×10^{-13} at a 0.1% physical error rate — the 'Teraquop' regime — co-designed for neutral-atom hardware. Note: this is a simulation result establishing a quantum-memory baseline, not an experimental hardware demonstration, and does not yet include fault-tolerant logical gates for universal computation. (postquantum.com — quera qldpc 2 to 1 physical logical qubit ratio, as of 2026-04-01, confidence: medium, source: ingest)
- *New large-scale qLDPC teraquop simulation result, distinct from the established IBM gross-code entries, with its simulation-only evidence status flagged per baseline rules.*
- Breakeven performance was demonstrated for qLDPC codes on a trapped-ion processor, with a code encoding 4 logical qubits into 18 physical qubits achieving logical error rates up to 9x better than prior superconducting demonstrations and qubit lifetimes comparable to or exceeding physical qubits. (Breakeven demonstration of quantum low-density parity-check codes, as of 2026-06-04, confidence: high, source: grok)
- *New qLDPC-specific breakeven result on trapped-ion hardware, distinct in code type and ratio from the existing Quantinuum concatenated-code and IBM gross-code entries.*

Hardware by modality

Trapped ion qubits (`trapped-ion-qubits`)

New developments (1)

University of Oxford

- University of Oxford demonstrated microwave-driven Mølmer-Sørensen two-qubit gates on 43Ca+ hyperfine qubits with durations of 154 μs (1.0(2)% error) and 331 μs (0.5(1)% error) in a cryogenic surface trap. (Robust and fast microwave-driven quantum logic for trapped-ion qubits, as of 2026-08-24, confidence: high, source: grok)
- *Reports a distinct microwave-driven gate approach and gate-speed/fidelity figures for Oxford not covered by the existing laser-driven, all-electronic, or Oxford Ionics EQC results.*

Superseded (2)

IonQ

- IonQ demonstrated electronically controlled two-qubit gates with estimated error of 8.4(7)×10^-5 (>99.99% fidelity) without ground-state cooling, remaining ≤5×10^-4 for average phonon occupation up to 9.4(3) on the gate mode. (Trapped-ion two-qubit gates with >99.99% fidelity without ground-state cooling, as of 2025-10-20, confidence: high, source: grok)
- supersedes: "[d0001:1] IonQ reported over 99.99% two-qubit gate fidelity on a trapped-ion R&D prototype using its Electronic Qubit Control (EQC) technology"
- *Provides the primary technical source and exact error figure behind the previously reported >99.99% EQC fidelity claim, adding the substantive new condition that fidelity holds without ground-state cooling across a range of phonon occupations.*

Quantinuum

- Quantinuum's Helios system is a 98-qubit trapped-ion QCCD processor using 137Ba+ hyperfine qubits with all-to-all connectivity via a rotatable ion storage ring and four-way junction, achieving average two-qubit gate infidelity of 7.9(2)×10^-4 (~99.921% fidelity), single-qubit infidelity of 2.5(1)×10^-5 (~99.9975% fidelity), and SPAM infidelity of 3.3(5)×10^-4. (A 98-qubit trapped-ion quantum computer with all-to-all connectivity, as of

2026-06-17, confidence: high, source: grok)
- supersedes: "[d0001:11] Reported figures for Quantinuum's Helios system (98 physical 137Ba+ qubits, November 2025) include an all-pairs two-qubit gate fidelity of 99.921% and a single-qubit gate fidelity of 99.9975%."
- *Upgrades the same figures from a medium-confidence secondary vendor-roundup citation to a peer-reviewed Nature publication with independent Sandia verification, adding architectural detail (rotatable ion storage ring, four-way junction) and a SPAM infidelity figure.*

Newly captured background -- older material, first time in the knowledge base (1)

AQT

- AQT's LYNX Series trapped-ion system achieved a quantum volume of 32768 with all-to-all connectivity in a 19-inch rack-mounted form factor operating at room temperature. (AQT Sets New European Industry Standard: Introducing the "LYNX" Series with Record-Breaking Quantum Volume, as of 2026-05-05, confidence: high, source: grok)
- *New system generation and metric for AQT (LYNX succeeding PINE) not previously present in the knowledge base.*

Neutral atom qubits (`neutral-atom-qubits`)

New developments (3)

Caltech

- Caltech Endres group published in Nature (August 2026) the first direct experimental measurement of 2D conformal field theory spectra using an analog neutral-atom quantum simulator with chains of up to 35 strontium atoms. (Caltech Researchers Measure Conformal Field Theory Spectra on a Neutral-Atom Quantum Simulator, as of 2026-08-20, confidence: high, source: grok)
- *Introduces a new analog-mode Caltech result (strontium chain CFT simulation) distinct from the existing digital cesium-array coverage.*

Infleqtion

- Japan's Institute for Molecular Science, with Hitachi and Infleqtion, launched "Shunkai," Japan's first full-stack neutral-atom quantum computer, starting at 50 qubits (target 10,000 by 2031), using neutral rubidium atoms and optical tweezers, operating at room temperature with reconfigurable qubit connectivity; cloud access planned, commercialization via Yaqumo Inc. (thequantuminsider.com — infleqtion japan operational neutral atom quantum computer, as of 2026-08-24, confidence: medium, source: ingest)
- *Reports a new device/deployment (Shunkai) not previously present in the base, distinct from the existing Sqale-platform coverage.*

Pasqal

- Pasqal had 7 neutral-atom QPUs with 100-200+ qubits deployed in commercial use and 3 more in production as of August 2026. (France's Pasqal is public, with $360M and a valuation of about 100 times revenue, as of 2026-08-28, confidence: high, source: grok)
- *New aggregate figure on the number and scale of Pasqal's deployed commercial QPUs, not previously recorded in this leaf.*

Newly captured background -- older material, first time in the knowledge base (3)

Harvard (Lukin group)

- Harvard Lukin group achieved 99.85% raw CZ gate fidelity (99.94% post-selected) in April 2026 results referenced in neutral-atom strategic plan. (Neutral-Atom Roadmap Charts a Path to Quantum Utility, as of 2026-04-01, confidence: medium, source: grok)
- *Provides a two-qubit Rydberg gate fidelity figure for Harvard not previously present in this leaf (existing Harvard entry covers only array size/loading, not gate fidelity).*

Institut d'Optique

- CNRS, the Institut d'Optique Graduate School, and Pasqal inaugurated the "AtomIQ-Lab" joint research laboratory on November 26, 2025, focused on laser-controlled cold-atom quantum computers and optimizing quantum-processor performance via atom-manipulation techniques. (pasqal.com — the cnrs linstitut doptique graduate school and the scaleup pasqal join forces to build the computers of the future, as of 2025-11-26, confidence: medium, source: ingest)
- *First coverage of Institut d'Optique in this leaf, introducing a new joint lab relevant to atom-manipulation performance.*

Pasqal

- Pasqal, with its subsidiary Aeponyx, demonstrated trapping individual atoms using laser light generated by a photonic integrated circuit (PIC), announced August 10, 2026 (vendor announcement) as a world-first, more scalable approach to qubit control aimed at future fault-tolerant processors. (globenewswire.com — Pasqal Brings Qubit Control On Chip Advancing the Path to Fault Tolerant Quantum Computing at Scale, as of 2026-08-10, confidence: medium, source: ingest)
- *Introduces a genuinely new atom-control technique (photonic-integrated-circuit-based trapping) not previously covered for Pasqal.*

Superconducting qubits (`superconducting-qubits`)

Superseded (2)

ETH Zurich

- ETH Zurich demonstrated a hybrid architecture coupling a superconducting transmon qubit to a high-overtone bulk-acoustic wave (HBAR) mechanical resonator, using it to perform C-PHASE gates and small quantum algorithms, published in Science. (Best of two worlds: Superconducting qubits and mechanical resonators for quantum computing, as of 2026-05-28, confidence: high, source: grok)
- supersedes: "[d0002:3] ETH Zurich researchers demonstrated a new quantum computer architecture in which superconducting qubits process information stored in microscopic mechanical resonators, using mechanical vibrations as quantum working memory."
- *Cites the primary ETH Zurich/Science source with concrete technical detail (HBAR resonator, C-PHASE gates, peer-reviewed publication) for the same result the base only knew via secondary commentary.*

Rigetti

- Rigetti's Q2 2026 report gives more precise figures for its 108-qubit Cepheus-1-108Q system: approximately 99.9% median single-qubit gate fidelity, approximately 99.1% median two-qubit gate fidelity, and approximately 60 nanosecond gate speeds. (sec.gov — rgti 20260806xex99d1, as of 2026-08-06, confidence: medium, source: ingest)
- supersedes: "[d0001:12] Rigetti reports median two-qubit gate fidelities of 99.7% on its 9-qubit system, 99.6% on its 36-qubit chiplet system, and 99% on its 108-qubit Cepheus-1-108Q system, together with 99.9% median one-qubit gate fidelity across its current product line."
- *A newer, more precise Q2 2026 filing updates the two-qubit fidelity figure for the same 108-qubit Cepheus-1-108Q system from 99% to 99.1% and adds a specific gate-speed figure.*

Newly captured background -- older material, first time in the knowledge base (5)

ETH Zurich

- ETH Zurich researchers demonstrated a new quantum computer architecture in which superconducting qubits process information stored in microscopic mechanical resonators, using mechanical vibrations as quantum working memory. (thequantuminsider.com — eth zurich demonstrates quantum computer architecture with mechanical working memory, as of 2026-07-09, confidence: medium, source: ingest)
- *First reported result for ETH Zurich in this leaf's knowledge base, describing a novel hybrid superconducting-mechanical qubit architecture.*

IBM

- IBM's `ibm_miami` — the first quantum processing unit built on the Nighthawk design — became available to customers on IBM's Premium and Flex cloud plans as of January 2026. (quantum.cloud.ibm.com — 2026 01 05 nighthawk, as of 2026-01-05, confidence: medium, source: ingest)
- *Reports a device-availability/deployment milestone for the Nighthawk-based ibm_miami system not present in the base, which covers only Nighthawk's design specs.*

IQM

- IQM deployed its first U.S. system, a 20-qubit Pathfinder Radiance device, at Oak Ridge National Laboratory in June 2026 for hybrid quantum-HPC integration. (IQM Deploys Its First U.S. Quantum Computer at Oak Ridge National Laboratory, as of 2026-06-17, confidence: high, source: grok)
- *Reports a new IQM deployment (first U.S. system, Oak Ridge) not present in the base, which covers only European deployments.*

MIT

- MIT researchers developed air-stable ultrathin NbSe2 superconductors via encapsulation epitaxy with graphene, enabling wafer-scale integration into superconducting microwave circuits. (Air-stable, ultrathin superconductors developed for more scalable quantum devices, as of 2026-08-05, confidence: high, source: grok)
- *First reported result for MIT in this leaf's knowledge base, describing a novel materials/fabrication advance for superconducting circuits.*

Yale

- Yale researchers fabricated all-nitride NbN/AlN/NbN transmon qubits via atomic layer deposition, achieving microsecond-scale T1 even above 300 mK. (All-nitride superconducting qubits based on atomic layer deposition, as of 2026-01-06, confidence: high, source: grok)
- *First reported result for Yale in this leaf's knowledge base, a peer-reviewed advance in transmon fabrication and coherence at elevated temperature.*

Spin qubits (`spin-qubits`)

New developments (1)

Intel

- Intel demonstrated tune-up of a 12-quantum-dot spin-qubit device (Tunnel Falls), the largest reported qubit count in a single spin-qubit device fabricated on a 300 mm semiconductor manufacturing line using EUV and HVM processes. (12-spin-qubit arrays fabricated on a 300 mm semiconductor manufacturing line, as of 2026-08-24, confidence: high, source: grok)

- *Introduces a specific qubit-count result (12-dot device) for Intel's 300-mm platform not previously recorded, which so far only reported uniformity/fidelity statistics without a device qubit-count figure.*

Newly captured background -- older material, first time in the knowledge base (3)

Delft (QuTech)

- QuTech researchers developed a new chip architecture — a 23×23 tile design — on a germanium/silicon-germanium (Ge/SiGe) heterostructure, with a stated design capacity to potentially host up to 1,058 hole-spin qubits; this is an announced architecture capacity, not a demonstrated qubit count or fidelity result. (eurekalert.org — 1116077, as of undated, confidence: low, source: ingest)
- *New tile-architecture capacity figure for QuTech's germanium platform, distinct from the existing Groove Quantum 18-qubit device, but flagged as an announced design capacity rather than an achieved qubit count.*
- QuTech demonstrated a planar 10-qubit germanium processor in 3-4-3 layout on Ge/SiGe heterostructure with uniform high-fidelity control in a two-dimensional layout where central qubits connect to four neighbours. (From Complexity to Control: a 10-spin qubit array in germanium - QuTech, as of 2025-11-27, confidence: high, source: grok)
- *Reports a distinct 2D 10-qubit germanium array architecture with 4-neighbour connectivity, separate from the previously recorded 18-qubit 2xN modular Groove Quantum array.*

RIKEN

- RIKEN demonstrated simultaneous high-fidelity single-qubit gates across a spin qubit array with primitive π/2 gate fidelities well above 99.99% for each qubit individually (some approaching 99.999%) using tailored control pulses on a shared line and pairwise phase calibrations. (Simultaneous High-Fidelity Single-Qubit Gates in a Spin Qubit Array, as of 2026-07-15, confidence: high, source: grok)
- *Provides specific quantitative fidelity figures (>99.99%, up to ~99.999%) for the RIKEN preprint that the existing d0001:27 entry described only qualitatively without numbers.*

Photonic quantum computing (`photonic-quantum`)

Newly captured background -- older material, first time in the knowledge base (3)

Quandela

- Quandela's Belenos 12-qubit photonic system was integrated into OVHcloud's Quantum Platform in April 2026 for cloud-based access to photonic quantum computing. (OVHcloud Expands Sovereign QaaS Platform via Quandela's Belenos Photonic System, as of 2026-06-17, confidence: high, source: grok)
- *New deployment/access channel for Belenos not previously recorded in the knowledge base.*
- Quandela demonstrated photonic quantum reservoir processing for machine learning and quantum tomography tasks on its Belenos QPU in June 2026 using a 24-mode programmable silicon photonic chip with quantum-dot single-photon sources. (Quandela Demonstrates Photonic Quantum Reservoir Processing for Advanced Machine Learning and Single-Basis Quantum Tomography, as of 2026-06-27, confidence: high, source: grok)
- *New application/architecture detail (24-mode chip, quantum-dot sources) not present in existing Quandela entries.*

USTC

- China Telecom Quantum Group launched the Tianyan-P2000 photonic quantum computer in June 2026, based on Jiuzhang 4.0 architecture, controlling 2,682 photons and connected to the Tianyan quantum cloud platform. (News, as of 2026-06-24, confidence: high, source: grok)
- *New commercial/deployment system derived from Jiuzhang 4.0 architecture, distinct from and not exceeding*

the 3,050-photon research result, so it adds rather than supersedes.